# Guideline: How to Find Use Cases

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**Key Principle**

Use cases are defined to satisfy the **goals of primary actors**. The process of discovering them involves clarifying the system boundary, identifying the primary actors and their goals, and defining use cases that fulfill these goals.

## Basic Procedure

1. **Choose the system boundary:** Decide whether the system includes just the software, hardware, external services, or an entire organization.
2. **Identify the primary actors:** Those who have goals achieved through the system's services.
3. **Identify goals for each primary actor:** What does each actor want to accomplish through the system?
4. **Define use cases:** Create use cases that fulfill these goals, naming them after the actor's goal (e.g., goal: "process sale" → use case: *Process Sale*).

*Note:* In iterative development, not all goals or use cases are discovered initially; they evolve over time.

## Step 1: Choose the System Boundary

Clarify what is inside and outside the system. For example, in the **POS (Point-of-Sale)** case study:

- The POS system itself is the system under design.
- External entities such as the cashier, payment authorization service, and customer are outside the boundary.

If unclear, identify external actors first—this helps define the boundary. For instance, since payment authorization is done externally, the payment service is an **external actor**.

# Steps 2 & 3: Find Primary Actors and Goals

In practice, actors and goals are often identified together during brainstorming sessions.

**Guideline**

Begin by brainstorming primary actors—it provides the framework for discovering user goals.

## Helpful Questions to Identify Actors and Goals

- Who starts or stops the system?
- Who performs system administration or user management?
- Does "time" trigger system actions (e.g., scheduled reports)?
- Is there a monitoring process that restarts the system after failure?
- Who evaluates logs, manages updates, or monitors performance?
- Are there external systems or robots interacting with this system?
- Who is notified of errors or system events?

# Organizing Actors and Goals

Two main approaches:

1. **Use Case Diagram:** Draw discovered actors and goals as use cases.
2. **Actor–Goal List:** List all actors and their corresponding goals, then refine and later represent in a diagram.

**Example Actor–Goal List:**

| Actor | Goals |
|---|---|
| Cashier | process sales, handle returns, cash in/out |
| System Administrator | add/modify/delete users, manage security |
| Manager | start up, shut down |
| Sales Activity System | analyze sales and performance data |

The *Sales Activity System* is a remote application that frequently requests data from POS nodes.

## Why Focus on Actor Goals?

Actors have goals and use systems to achieve them. Use case modeling begins by finding actors and their goals, not tasks or steps.

**Prefer asking:**

- "Who uses the system, and what are their goals?"

**Over asking:**

- "What do you do?" (which often reflects current tasks or procedures)

**Benefit:** Goal-oriented questions open up the vision for new and improved solutions, increase business value, and align requirements with stakeholder intentions.

## Is the Cashier or Customer the Primary Actor?

The answer depends on the chosen **system boundary:**

- In a full *checkout service* context, the **customer** is the primary actor (goal: buy goods and leave).
- In the *POS system* boundary, the **cashier** is the primary actor (goal: process sale efficiently).

Thus, in the **NextGen POS system**, the cashier—not the customer—is primary because:

- The system is optimized for trained cashier use.
- The customer cannot effectively operate the POS terminal.

*However,* in systems like ticket-booking websites usable by both customers and agents, the **customer** would be the true primary actor.

## Event Analysis Approach

Another way to discover actors and goals is by examining **external events** that the system must respond to.

**Example:**

| External Event | From Actor | Goal / Use Case |
|---|---|---|
| Enter sale line item | Cashier | Process a Sale |
| Enter payment | Cashier or Customer | Process a Sale |

Grouping events by purpose helps reveal related use cases.

## Step 4: Define Use Cases

In general, define one use case per user goal, naming it after the goal.

**Example**

Goal: Process a Sale

Use Case: *Process Sale*

**Exception:** For CRUD-style operations (Create, Retrieve, Update, Delete), combine goals into a single use case, e.g., *Manage Users*.

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*Summary: Identify actors, discover their goals, and define use cases that deliver observable value within a well-defined system boundary.*